

Chase Christenson

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Summary

Data driven, quantitative researcher with 7+ years of experience in an academic setting, looking for opportunities to make a real-world impact through computational modeling. I'm excited about finding new challenges to address with applied mathematics. My research is focused on constructing mathematical models of cancer and developing new computational tools for improving data analysis.

Professional Experience

Postdoctoral Researcher **Jan 2026 – Present**
Oncology Center-Quantitative Sciences – *The Johns Hopkins University*

- Developing spatial models of cancer growth and response to therapy with agent-based models, coupled to whole-body quantitative systems pharmacology
- Developing methods for parameterizing computational models from multi-modal data sources, including single cell RNA sequencing, spatial transcriptomics, and immunohistology

Postdoctoral Researcher **Jan 2025 – Dec 2025**
Scientific Computing and Imaging Institute – *University of Utah*

- Developed multiscale model of melanoma growth combining principles from systems biology and cellular growth and response to therapy.
- Developed and improved upon scientific machine learning algorithms for identification of non-linear systems from data.
- Utilized differentiable programming techniques to improve methods for optimal control of non-convex, constrained problems.

Graduate Research Assistant **Aug 2020 – Dec 2024**
Center for Computational Oncology – *University of Texas at Austin*

- Developed imaging processing pipelines for 3D medical image datasets consisting of CT, SPECT, and/or MRI; used segmentation methods, rigid and non-rigid registration, data normalization and mathematical formulations to prepare images for visualization and analysis.
- Constructed frameworks in MATLAB and Python for analysis of longitudinal imaging data with mechanism based models. These works required writing finite difference code from scratch to implement variations on the Fisher-Kolmogorov equation. Parameters were estimated within the frameworks using either a least-squares minimization with custom gradient solvers, or Bayesian methods for introducing uncertainty.
- Developed a framework for building projection based reduced order model directly from processed imaging data. The custom, intrusive solver was implemented in both MATLAB and Python and is capable of simulating the 3D growth and response of breast cancer up to 3 orders of magnitude faster than the full model, with similar accuracy.
- Constructed digital twins capable of providing early predictions of response to therapy for glioblastoma and breast cancer patients. Constrained optimization problems based on clinical goals can be solved using a calibrated twin to provide patient specific feedback on therapeutic regimens with derivative free methods.

Undergraduate Research Assistant **May 2018 – May 2020**
Advanced Biophotonics and Nanomaterials Laboratory – *University of Texas at San Antonio*

- Worked in laboratory setting, evaluating protocols for chemical binding and detection abilities of a novel biosensor; custom engineered pneumatic system for delivering flow through microfluidic channels.
- Developed MATLAB programs for repeatable analysis of image outputs from the biosensor. Time dependent signals were determined from longitudinal images based on object recognition and used to estimate binding with the biosensor.

Education

- Ph.D. in Biomedical Engineering – 3.79 GPA** **2020 - 2024**
University of Texas at Austin
- Advisor: Thomas Yankeelov, Ph.D.
 - Concentration in computational biomedical engineering
- M.S. in Biomedical Engineering – 3.79 GPA** **2020 - 2022**
University of Texas at Austin
- Bachelor of Science in Biomedical Engineering – 3.96 GPA** **2017 - 2020**
University of Texas at San Antonio
- Concentration in Bio-imaging
 - Minor in Computer Science

Peer Reviewed Publications

- Christenson, C, Viknesh, S, Judson-Torres, RL, Arzani, A, Data-driven system identification in cancer systems biology: A multiscale modeling approach to melanoma, *Computer Methods and Programs in Biomedicine* 285, (2026), <https://doi.org/10.1016/j.cmpb.2026.109511>
- Patel, RJS, Das, A, Anil, A, Patel, G, Woodall, RT, Christenson, C, Thiagarajan, T, Lima, EABF, Lorenzo, G, Hormuth, DA, Rockne, RC, Adhikarla, V, Wu, C, Yankeelov, TE. *Integrating imaging and mathematical modeling to predict and optimize patient outcomes in oncology*, *npj Systems Biology and Applications*, (2026), <https://doi.org/10.1038/s41540-026-00764-6>
- Viknesh, S, Tatari, Y, Christenson, C, Arzani, A. *ADAM-SINDy: An Efficient Optimization Framework for Parameterized Nonlinear Dynamical System Identification*, *arXiv*, (2025), <https://doi.org/10.48550/arXiv.2410.16528>
- Christenson, C, Wu, C, Hormuth, DA, Ma, J, Yam, C, Rauch, GM, Yankeelov, TE. *Personalizing neoadjuvant chemotherapy regimens for triple-negative breast cancer using a biology-based digital twin*, *npj Systems Biology and Applications*, 11 (2025), 1-16. <https://doi.org/10.1038/s41540-025-00531-z>
- Christenson, C, Wu, C, Hormuth, DA, Stowers, CE, LaMonica, M, Ma, J, Rauch, GM, Yankeelov, TE. *Fast model calibration for predicting the response of breast cancer to chemotherapy using proper orthogonal decomposition*, *Journal of Computational Science*, 82 (2024), <https://doi.org/10.1016/j.jocs.2024.102400>
- Wu, C, Hormuth, DA, Christenson, C, Woodall, RT, Abdelmalik, MRA, Phillips, WT, Hughes, TJR, Brenner, AJ, Yankeelov, TE. *Image-guided patient-specific optimization of catheter placement for convection-enhanced nanoparticle delivery in recurrent glioblastoma*, *Computers in Biology and Medicine*, 179 (2024), 108889–. <https://doi.org/10.1016/j.compbiomed.2024.108889>
- Christenson, C, Wu, C, Hormuth, DA, Huang, S, Bao, A, Brenner, A, Yankeelov, TE. *Predicting the spatio-temporal response of recurrent glioblastoma treated with rhenium-186 labelled nanoliposomes*, *Brain Multiphysics*, 5 (2023), 100084–. <https://doi.org/10.1016/j.brain.2023.100084>
- Hormuth, DA, Farhat, M, Christenson, C, Curl, B, Quarles, C, Chung, C, Yankeelov, TE. *Opportunities for improving brain cancer treatment outcomes through imaging-based mathematical modeling of the delivery of radiotherapy and immunotherapy*, *Adv. Drug Deliv. Rev.*, 187 (2022), 114367. <https://doi.org/10.1016/j.addr.2022.114367>
- Holland, MD, Morales, A, Simmons, S, Smith, B, Misko, SR, Jiang, X, Hormuth, DA, Christenson, C, Koomullil, RP, Morgan, DE, Li, Y, Xu, J, Yankeelov, TE, Kim, H., *Disposable point-of-care portable perfusion phantom for quantitative DCE-MRI*, *Med Phys.*, 49(1) (2022), 271-81. <https://doi.org/10.1002/mp.15372>

Conferences

- Christenson, C, Wu, C, Hormuth II, DA, Pash, G, Stowers, C, LaMonica, M, Willcox, K, Ma, J, Rauch, GM, Yankeelov, TE. *Introducing mechanical stress into a reduced order model for breast cancer response to chemotherapy* [Poster abstract]. 2023 Biomedical Engineering Society annual meeting. (October 13, 2023).
- Christenson, C, Wu, C, Hormuth II, DA, Pash, G, Stowers, C, LaMonica, M, Willcox, K, Ma, J, Rauch, GM, Yankeelov, TE. *Fast digital twin construction for modeling the response of breast cancer to therapy using*

proper orthogonal decomposition [Invited presentation abstract]. 2023 Society for Mathematical Biology meeting. (July 9, 2023).

Christenson, C, Wu, C., Hormuth II, D.A., Huang, S., Brenner, A., Yankeelov, T.E., *A biology-based, mathematical model to predict the response of recurrent glioblastoma to treatment with ¹⁸⁶Re-labeled nanoliposomes* [Poster abstract]. American Association of Cancer Research annual meeting. (2022).

Wu, C, Hormuth, D, Christenson, C, Abdelmalik, M, Phillips, W, Hughes, T, Brenner, A, and Thomas, Y. *RADT-14. TOWARDS IMAGE-GUIDED MODELING OF PATIENT-SPECIFIC RHENIUM-186 NANOLIPOSOME DISTRIBUTION VIA CONVECTION-ENHANCED DELIVERY FOR MULTIFORME*, Neuro-oncology (Charlottesville, Va.) 23.Supplement_6 (2021).

Christenson, C., Baryeh, K., Ahadian, S., Nasiri, R., Dokmeci, MR, Goudie, M, Yong Ye J, et al., *Enhancement of label-free biosensing of cardiac troponin I*. In: Label-Free Biomedical Imaging and Sensing (LBIS) (Vol. 11251, p. 112512J). International Society for Optics and Photonics. (2020).
<https://doi.org/10.1117/12.2546979>

Christenson, C., Baryeh, K., Ahadian, S., Nasiri, R., Dokmeci, M. R., Goudie, M., Khademhosseini, A., Yong Ye, J. *Optimization of PC-TIR Sensor for Cardiac Troponin I Detection using Protein-G* [Poster Abstract]. UTSA Undergraduate Research Symposium. (2019).

Awards

- 1st Place in UTSA CITE \$100K Student Technology Venture Competition
- Co-Winner of Highest Academic Achievement Award in BME, UTSA Class of 2020
- President's List – Spring 2017, Fall 2017, Spring 2018, Fall 2018
- Dean's List – Spring 2019

Skills

Programming languages: MATLAB | Python | C | C++ | CUDA | LabView | LaTeX

Computational: numerical simulations | parameter estimation | predictive modeling | image processing | optimal control | digital twins | Bayesian inference | statistical analysis | parallel computing | batch programming | GitHub | model selection | cluster analysis | decision trees | convolutional neural networks | reduced order modeling | linear/log regression | PyTorch | system identification | differentiable programming | agent-based modeling

Biology: tumor biology | interpreting *in vivo* imaging data (MRI, SPECT/PET, CT) | quantitative systems biology

General: peer-reviewed publication | technical writing | presentations | project management | problem solving

References

Thomas E. Yankeelov, Professor, Director of Center for Computational Oncology (Dissertation Advisor)

Department of Biomedical Engineering

University of Texas at Austin

thomas.yankeelov@utexas.edu

Amirhossein Arzani, Associate Professor (Postdoc Advisor)

Department of Mechanical Engineering & Scientific Computing and Imaging Institute

University of Utah

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Atul Deshpande, Assistant Professor (Postdoc Advisor)

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